DEVELOPMENT PC / 1 OF 4

Create and transfer the ZIP

Do this on your development PC first. The client Print PC does not build the release.

A. Use the approved source

Use the latest approved main source containing Factory Agent v1.2.1. On this development PC, open PowerShell in the project folder:

```
cd D:\LuckyBags\Automation\print-production-automation

.\automation\make-release.ps1
```

If your project is stored elsewhere, change only the cd path. The command creates the release and checksum; it does not install the agent or send print jobs.

B. Find the two generated files

Open the project's **automation\dist** folder. Keep these files together:

```
LuckyBags-Factory-Agent-v1.2.1.zip

LuckyBags-Factory-Agent-v1.2.1.sha256.txt
```

Do not ZIP the entire automation folder manually. The release script includes only approved release files and excludes .env, credentials, developer environments, logs and downloaded artwork.

C. Transfer and verify on the Print PC

Transfer both files using your trusted method, for example a controlled USB drive. If they are saved under C:\Downloads on the Print PC, run:

```
Get-FileHash `
  "C:\Downloads\LuckyBags-Factory-Agent-v1.2.1.zip" `
  -Algorithm SHA256
Get-Content `
  "C:\Downloads\LuckyBags-Factory-Agent-v1.2.1.sha256.txt"
```

Replace C:\Downloads with the actual transfer folder. Compare the hash returned by Get-FileHash with the hash in the text file. Letter case does not matter, but every hexadecimal character must match. **Stop if they differ.** Obtain the expected hash through a trusted channel; a matching checksum alone does not prove the publisher is trusted.

D. Scan, then extract

On the Print PC, scan the ZIP with Microsoft Defender before extraction. Extract directly to **C:\LuckyBagsFactory** and continue on page 2. Keep Auto Print OFF throughout the pilot.

Rebuilding the ZIP can change its checksum. Always use the checksum generated alongside that exact ZIP. The checksum on page 4 identifies the reviewed ZIP originally supplied with this guide.

Prepare and install

For a controlled Windows pilot. Read all four pages before using a live order.

Keep Epson Edge Print Auto Print OFF. Queueing on the website does not immediately start physical printing.

1. Deploy the web update first

Web administrator: deploy the latest approved main release and activate it using the existing server deployment procedure. Back up the database first. Do not reset or seed production data.

2. Prepare the client PC

Use patched Windows, current Microsoft Defender, genuine Python 3.13 x64 with security updates, and a dedicated **standard operator account**. Confirm the actual Epson Edge Print hot folder. Do not run the agent as Administrator or SYSTEM.

3. Extract and install

Verify and scan **LuckyBags-Factory-Agent-v1.2.1.zip**, then extract directly into **C:\LuckyBagsFactory**. IT may need to create that folder and grant the operator access. Open ordinary PowerShell there:

```
cd C:\LuckyBagsFactory

.\install.ps1 -PythonExe "C:\Actual\Path\To\Python313\python.exe"
```

Replace the interpreter path with the signed Python installation on this PC. If scripts are blocked, ask IT to approve them; do not bypass execution policy.

4. Configure .env and credentials

Edit **C:\LuckyBagsFactory\.env**. Never include the real key or credential JSON in screenshots, email or release ZIPs.

Setting	What to enter
PRODUCTION_API	https://lbprint.app.keenpillar.com/api
FACTORY_API_KEY	Unique secret matching the web server
GOOGLE_SERVICE_ACCOUNT_FILE	Absolute path to a dedicated Google Viewer JSON
FACTORY_PRINT_DIR	C:\LuckyBagsFactory\factory_print
EDGE_PRINT_HOT_FOLDER	Actual existing Epson pilot hot folder
REQUIRE_DEFENDER_SCAN / ALLOW_LOCAL_HTTP	true / false

Use a dedicated Google account with Viewer access only to approved artwork. Do not copy the web application's Editor credential to the Print PC.

Run one controlled pilot

Use a disposable order with quantity one. Test single-sided first, then double-sided.

5. Check the local installation

Run these commands as the dedicated operator:

```
.\protect-agent.ps1

.\run-once.ps1 -CheckOnly
```

Stop if either command fails. A successful CheckOnly verifies configuration, not Google/API connectivity, Defender scanning or printer output.

6. Approve and queue the right artwork

On the website, compare the images and approve the **right-side Drive originals**. Single-sided products use one PRINT file; double-sided products use both FRONT and BACK files. The Shopify mockup on the left is not sent for printing. Open Ready for Print and create the locked batch.

7. Prepare the queued batch

On the Print PC, run:

```
.\run-once.ps1
```

Confirm the website batch becomes **PREPARED**. Inspect the workspace manifest: it must contain the expected one or two artwork files and correct order quantities. Nothing should physically print at this stage.

8. Authorize delivery to Epson Edge Print

On the batch page, review the exact batch and sign the **HANDOFF** confirmation. Run the same command again:

```
.\run-once.ps1
```

The agent verifies/scans the files and delivers them to the configured hot folder. Confirm **SENT_TO_RIP** in the website. With Auto Print OFF, check the jobs in Epson Edge Print: artwork, scale, media/profile, side pairing, layout and copies.

9. Print and confirm completion

Start physical printing manually in Epson Edge Print. Inspect the finished output. Only then sign the **PRINTED** confirmation on the website. Repeat the pilot with a double-sided order before regular use.

Quantity limit: automatic hot-folder handoff is blocked when any order quantity is above one. Copy/layout handling has not been validated for multi-set production. Do not bypass this by changing order quantities.

After both pilots pass, enable logon polling with **.\install-autostart.ps1**. Keep the operator signed in and use only one agent PC for this queue.

Know when to stop

Keep this page near the Print PC for the first pilot and subsequent updates.

What you see	What to do
No queued batch is prepared	Check the website batch is QUEUED, the agent is running, the API URL/key match, and the Google Viewer account can read the originals.
PREPARED but no Edge Print job	Complete the website HANDOFF review, then run the agent again. Check the configured hot folder and the real Epson preset.
FAILED, timeout or interrupted handoff	Stop. Inspect the hot folder, Edge Print queue and physical output. Do not delete manifests/journals or blindly requeue: some files may already have been delivered.
Defender, checksum or revision failure	Keep the batch stopped. Investigate the file and obtain a trusted original. Do not disable scanning or rename a file to bypass the check.
More than one set is required	Use an explicitly managed manual workflow, or wait for a validated quantity adapter. Automatic multi-set delivery is blocked.

Security checklist

Keep Windows, Defender, Python, Epson software and drivers updated. Protect the installation, credentials and hot folder with Windows permissions. Do not expose the hot folder publicly, run multiple agents or add antivirus exclusions for artwork. Stop the agent task before updates.

Windows administrators and the agent account can still stop or remove local software. The website cannot make the folder undeletable. This agent has no inbound service or arbitrary remote shell/deletion facility. A ZIP and checksum provide packaging/integrity, not protection against a compromised PC.

Release verification

Version 1.2.1: 17 isolated agent tests passed; the release ZIP received a local Defender scan; OSV returned no known advisory matches for 23 pinned packages on 2 November 2026. No client-PC or physical-printer test was performed. These checks are not a malware-free guarantee.

Expected ZIP SHA256 (join the two lines without spaces):

```
5544343af40a866c1a1a7d76ae21d3bc
af2c8d5380dca02d9e952dbd915f7260
```

New web batches carry available Drive checksum/revision evidence. Older batches may lack it; create a fresh approved pilot batch after deploying the web update. Full instructions and recovery details are in README.md and SECURITY_REVIEW.md inside the ZIP.